Set Operations

PreCalculus-I

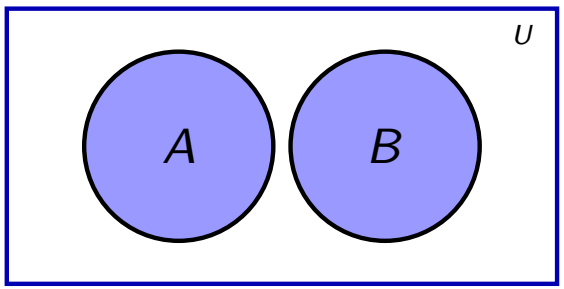
AH Sheikh

26CS (Evening) - Fall 2026

Union and Intersection

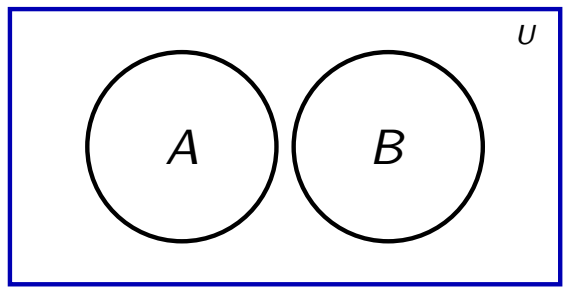
Union: elements in A, B, or both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



Intersection: elements in both A and B.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



Examples:

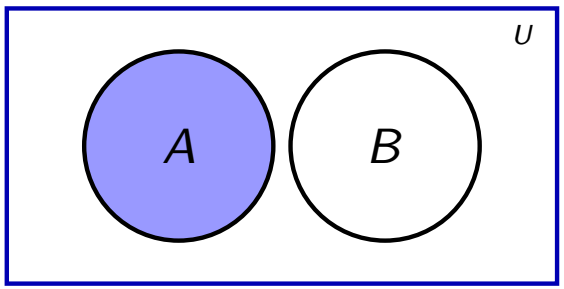
$A = \{1, 2, 3\}, B = \{2, 4, 6, 8\} \Rightarrow A \cup B = \{1, 2, 3, 4, 6, 8\},\ A \cap B = \{2\}.$

$A =$ evens, $B =$ odds $\Rightarrow A \cup B = \mathbb{Z},\ A \cap B = \emptyset$ (disjoint).

Difference and Complement

Difference: in A but not in B.

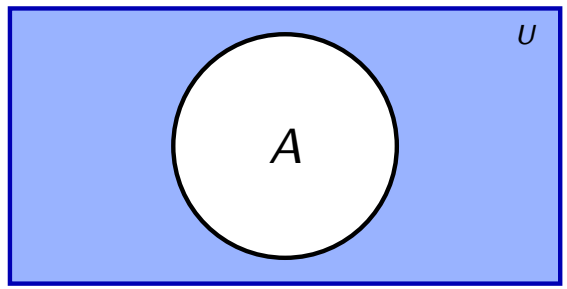
$$A - B = \{x \mid x \in A \wedge x \notin B\}$$



Ex: $A = \{1, 2, 3\}, B = \{2, 4\} \Rightarrow$
$A - B = \{1, 3\}$.

Complement: $U - A$.

$$\overline{A} = \{x \mid x \in U \wedge x \notin A\}$$



Ex: $A = \{a, b, c, d\}, U = \text{alphabet} \Rightarrow$
$\overline{A} = \{e, f, \ldots, z\}$.

Cardinality of Union & Summary

Cardinality of union: For $A = \{1, 2, 3\}, B = \{2, 3, 4\}$, $|A| = |B| = 3$, $|A \cup B| = 4$.

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Operation	**Notation**
Union	$A \cup B = \{x \mid x \in A \vee x \in B\}$
Intersection	$A \cap B = \{x \mid x \in A \wedge x \in B\}$
Difference	$A - B = \{x \mid x \in A \wedge x \notin B\}$
Complement	$\overline{A} = U - A = \{x \mid x \in U \wedge x \notin A\}$

Disjoint sets: $A \cap B = \emptyset$ (e.g. evens and odds in $\mathbb{Z}$).

Set Identities: Basic Laws

Identity Laws

$$A \cup \emptyset = A, \quad A \cap U = A$$

Domination Laws

$$A \cup U = U, \quad A \cap \emptyset = \emptyset$$

Idempotent Laws

$$A \cup A = A, \quad A \cap A = A$$

Complementation Law

$$\overline{(\overline{A})} = A$$

Complement Laws

$$A \cup \overline{A} = U, \quad A \cap \overline{A} = \emptyset$$

Commutative Laws

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

Associative: $A \cup (B \cup C) = (A \cup B) \cup C$, $A \cap (B \cap C) = (A \cap B) \cap C$.
Absorption: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$.

De Morgan's and Distributive Laws

De Morgan's Laws

$$\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}, \qquad \boxed{\overline{A \cap B} = \overline{A} \cup \overline{B}}$$

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Worked example:

$$\begin{aligned} \overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{(B \cap C)} && \text{(De Morgan)} \\ &= \overline{A} \cap (\overline{B} \cup \overline{C}) && \text{(De Morgan)} \end{aligned}$$

Four Methods for Proving Set Identities

1. Use the basic set identities
2. Use membership tables
3. Prove each set is a subset of the other
4. Use set builder notation and logical equivalences

The "subset of each other" method is often the most rigorous for elementary proofs.

Membership Tables

1 = element belongs; 0 = element does not.

Basic table:

A	B	$A \cup B$	$A \cap B$	$A - B$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	0
0	0	0	0	0

Distributive law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Columns 5 and 8 match $\Rightarrow$ identity holds.

Proof by Subset Inclusion

Goal: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Part 1: $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$. Assume $x \in A \cap (B \cup C)$.

$$
\begin{aligned}
&\Rightarrow (x \in A) \wedge (x \in B \cup C) && \text{(def. } \cap\text{)} \\
&\Rightarrow (x \in A) \wedge (x \in B \vee x \in C) && \text{(def. } \cup\text{)} \\
&\Rightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) && \text{(distributive)} \\
&\Rightarrow (x \in A \cap B) \vee (x \in A \cap C) && \text{(def. } \cap\text{)} \\
&\Rightarrow x \in (A \cap B) \cup (A \cap C) && \text{(def. } \cup\text{)}
\end{aligned}
$$

Part 2: $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. Assume $x \in (A \cap B) \cup (A \cap C)$.

$$
\begin{aligned}
&\Rightarrow (x \in A \cap B) \vee (x \in A \cap C) \\
&\Rightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\
&\Rightarrow x \in A \wedge (x \in B \vee x \in C) \\
&\Rightarrow x \in A \wedge (x \in B \cup C) \\
&\Rightarrow x \in A \cap (B \cup C)
\end{aligned}
$$

Both directions hold $\Rightarrow$ $\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$.

Proof via Set Builder Notation

Example: Show $\overline{A \cap B} = \overline{A} \cup \overline{B}$.

$$
\begin{aligned}
\overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{(def. complement)} \\
&= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{(def. } \in \text{ for } \cap) \\
&= \{x \mid x \notin A \vee x \notin B\} && \text{(De Morgan for logic)} \\
&= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{(def. complement)} \\
&= \overline{A} \cup \overline{B} && \text{(def. union)}
\end{aligned}
$$

$$\boxed{\overline{A \cap B} = \overline{A} \cup \overline{B}}$$

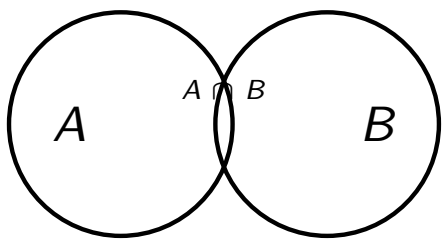
Inclusion–Exclusion Principle

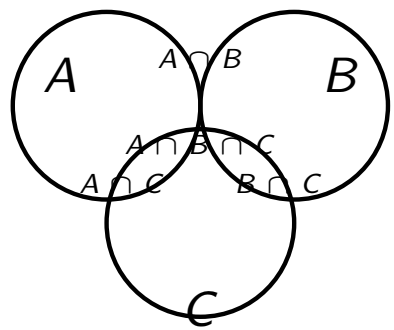
Two sets:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Three sets:

$$\boxed{|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|}$$





Worked Example: Two Lists

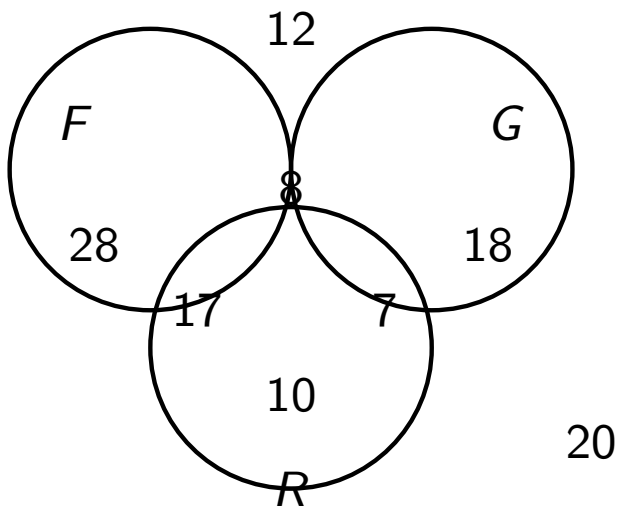
List A: 30 students in mathematics. List B: 35 students in English. Both lists: 20 names.

- Only on A: $30 - 20 = 10$
- Only on B: $35 - 20 = 15$
- On A or B (or both): $|A \cup B| = 30 + 35 - 20 = 45$
- On exactly one list: $10 + 15 = 25 = |A \oplus B|$

"Exactly one list" uses symmetric difference $A \oplus B = (A \cup B) - (A \cap B)$.

Worked Example: Three Languages

120 students. F: 65, G: 45, R: 42. $F \cap G$: 20, $F \cap R$: 25, $G \cap R$: 15. $F \cap G \cap R$: 8.



Exactly one course: $28 + 18 + 10 = 56$.

Summary

Concept	**Key Formula**
Union	$A \cup B = \{x \mid x \in A \vee x \in B\}$
Intersection	$A \cap B = \{x \mid x \in A \wedge x \in B\}$
Difference	$A - B = \{x \mid x \in A \wedge x \notin B\}$
Complement	$\overline{A} = \{x \mid x \in U \wedge x \notin A\}$
De Morgan	$\overline{A \cup B} = \overline{A} \cap \overline{B}, \ \ \overline{A \cap B} = \overline{A} \cup \overline{B}$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Inclusion–Exclusion (2)	$\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert$
Inclusion–Exclusion (3)	$\lvert A \cup B \cup C \rvert = \lvert A \rvert + \lvert B \rvert + \lvert C \rvert - \lvert A \cap B \rvert - \lvert A \cap C \rvert - \lvert B \cap C \rvert + \lvert A \cap B \cap C \rvert$

Practice Problems

1. $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 5, 7\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.
2. $U = \{1, \ldots, 10\}$, $A = \{1, 2, 3, 4\}$. Find $\overline{A}$.
3. Are $A = \{\text{even integers}\}$ and $B = \{\text{multiples of } 4\}$ disjoint? Justify.
4. Prove or disprove: $A - (B \cup C) = (A - B) \cap (A - C)$.
5. Verify with a membership table: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
6. 60 students: 30 like tea, 25 coffee, 10 both. How many like neither?
7. 200 people: 90 English, 80 French, 60 German; $E \cap F$: 30, $E \cap G$: 20, $F \cap G$: 15; all three: 5. How many speak none?